Mastering R Basics

Key Terms, Core Functions & Essential Packages

Ethiopian Statistics Service (ESS)

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Overview & Core Foundations

Training Foundation

Mastering R requires a solid foundation in core data structures, key data manipulation verbs, basic analytical functions, and the grammar of graphics.

- **1. Data Structures**: Vectors, Data Frames, Factors, Lists
- **2. Manipulation**: Filtering, transforming, aggregating
- **3. Analysis**: Summary statistics & basic testing
- **4. Visualization**: Layered graphics with `ggplot2`

Core R Concepts & Data Structures

Vector

Fundamental 1D container holding elements of the same data type.

- **Functions**: `c()`, `seq()`, `rep()`, `length()`, `class()`

Data Frame

2D tabular structure where columns hold different data types.

- **Functions**: `data.frame()`, `str()`, `dim()`, `head()`, `tail()`, `names()`

Factor

Vector type used for categorical data with defined levels.

- **Functions**: `factor()`, `levels()`, `as.factor()`

List

Flexible object holding mixed types, vectors, or nested lists.

- **Functions**: `list()`, `str()`

Data Manipulation & Wrangling (`dplyr` & `tidyr`)

- **Pipe Operator (|> or %>%)**: Passes output directly into the next function.
- **Selection & Filtering**: Subsetting variables or observations.
  - **Functions**: `select()` (columns), `filter()` (rows)
- **Transformation**: Modifying existing columns or creating new calculated fields.
  - **Functions**: `mutate()`, `rename()`, `arrange()`
- **Aggregation**: Grouping data to compute group-level summaries.
  - **Functions**: `group_by()`, `summarize()`, `count()`
- **Reshaping**: Converting datasets between wide and long formats.
  - **Functions**: `pivot_longer()`, `pivot_wider()`

Data Analysis & Summary Statistics

Descriptive Statistics

Calculating central tendency, dispersion, and overall distribution.

- **Key Functions**: `mean()`, `median()`, `sd()`, `quantile()`, `summary()`

Frequency & Cross-Tabulation

Counting categorical counts and relative proportions.

- **Key Functions**: `table()`, `prop.table()`

Basic Statistical Tests

Examining correlations and basic linear relationships.

- **Key Functions**: `cor()`, `t.test()`, `lm()`

Data Visualization (ggplot2)

- **Aesthetics (aes)**: Mapping dataset variables to visual dimensions (x, y, color, fill, size).
- **Geometries (geom_*)**: Declaring geometric objects:
  - `geom_point()`: Scatter plots | `geom_bar()`: Frequency counts
  - `geom_histogram()`: Continuous distributions
  - `geom_boxplot()`: Outliers / IQR | `geom_line()`: Trend lines
- **Faceting**: Splitting graphics into multiple subplots by category.
  - **Key Functions**: `facet_wrap()`, `facet_grid()`

Essential Package Ecosystem

Package	Primary Role
`tidyverse`	Master library bundling `dplyr`, `ggplot2`, `tidyr`, `readr`, and `stringr`
`readr` / `readxl`	Reading CSV and Excel spreadsheets into clean data frames
`dplyr`	Core grammar for fast data manipulation
`tidyr`	Functions for data cleaning and reshaping
`ggplot2`	Layered visualization framework